

LAHORE GARRISON UNIVERSITY

Department of Computer Science

Computer Networks (Lab)



SEMESTER PROJECT REPORT

An Integrated and Secure Network Topology for
Al-Shifa Smart Hospital

Student Name	Rauf Hassan
Roll Number	FA-2023/BSCS/093
Section	C
Course	Computer Networks (Lab)
Submitted To	Miss Hafsa Tahir
Date	May 2025
Session	2023 – 2027

Table of Contents

Abstract.....	3
1. Introduction & Problem Statement.....	3
1.1 Background.....	3
1.2 Deficiencies of the Original Flat Network.....	3
1.3 Project Objectives.....	4
2. Network Design Methodology.....	4
2.1 Hierarchical Three-Tier Architecture.....	4
2.2 Redundancy and High Availability.....	5
3. Implementation & Working.....	6
3.1 VLAN Segmentation.....	6
3.2 VLSM IP Addressing Scheme.....	7
3.3 Routing Strategy.....	8
3.4 Network Services.....	9
3.5 Security & Hardening.....	10
3.6 QoS for VoIP.....	12
4. Testing & Verification.....	13
4.1 DHCP and Inter-VLAN Connectivity.....	14
4.2 OSPF Routing and WAN Connectivity.....	14
4.3 Guest VLAN ACL Isolation.....	15
4.4 NAT Translation Verification.....	15
4.5 Port Security Violation.....	15
5. Challenges & Limitations.....	16
5.1 Implementation Challenges.....	16
5.2 Inherent Limitations of Cisco Packet Tracer.....	17
6. Future Aspects & Enhancements.....	18
6.1 Dedicated Next-Generation Firewall and IDS/IPS.....	18
6.2 IPv6 Dual-Stack Migration.....	18
6.3 SD-WAN Replacement for Serial WAN.....	18
6.4 IPsec VPN for Encrypted WAN Transit.....	18
6.5 Network Access Control (NAC) and 802.1X Authentication.....	18
7. Conclusion.....	19
References.....	19

Abstract

Al-Shifa Smart Hospital — a fictional 600-bed tertiary-care facility previously operated on a flat, unmanaged network in which every clinical department, administrative workstation, and internet-facing system shared a single broadcast domain. This architecture exposed sensitive Electronic Health Records (EHRs) and PACS radiology images to trivial interception, introduced broadcast storm risk across hundreds of networked medical devices, and provided no prioritization for life-critical VoIP and telemetry traffic.

This project addresses those deficiencies by designing and fully implementing a HIPAA-aligned, three-tier hierarchical network topology within Cisco Packet Tracer 8.x. The solution deploys twelve isolated VLANs across the VLSM-subnetted 192.168.10.0/24 address space, inter-VLAN routing via Switched Virtual Interfaces (SVIs) on redundant Core Multilayer Switches, and OSPF Area 0 dynamic routing connecting the main campus to a remote Branch Clinic over a simulated serial WAN link. Layered security is enforced through Extended ACLs (GUEST_ISOLATION and BRANCH_SECURITY), NAT Overload (PAT) for internet access, and Layer-2 hardening via Port Security with sticky MAC learning, PortFast, and BPDU Guard on all access ports. Quality of Service for VoIP is implemented through a dedicated Voice VLAN (110) with MLS QoS COS trust boundaries. The finished Packet Tracer build was systematically verified through connectivity, routing, ACL isolation, NAT translation, and port-security violation tests, all returning expected outcomes. The project demonstrates a comprehensive translation of the initial proposal into a working enterprise-grade hospital network simulation.

1. Introduction & Problem Statement

1.1 Background

Modern hospitals are among the most network-intensive environments in any industry. Patient Electronic Health Records (EHRs), real-time telemetry from ICU monitoring equipment, PACS systems streaming high-resolution radiology images, VoIP communications between clinical staff, and administrative management systems all run simultaneously and interdependently on the same underlying network fabric. Unlike a corporate environment, a network failure or security breach in a hospital is not merely a business inconvenience — it can directly endanger patient lives.

This project was undertaken to redesign the network of Al-Shifa Smart Hospital, a fictional 600-bed tertiary-care facility used as the simulation scenario. The objective was to replace the hospital's existing flat network topology with a structured, secure, and redundant architecture built to the same standards applied in real-world healthcare IT deployments.

1.2 Deficiencies of the Original Flat Network

The pre-existing network at Al-Shifa placed all departments, the Intensive Care Unit, Emergency Ward, Radiology, Pharmacy, Administration, and Server Room within a single broadcast domain. This design introduced the following critical deficiencies:

- **Security Vulnerability:** All traffic, including unencrypted EHR transactions and PACS image transfers, traversed the same broadcast domain. Any device on the network could trivially intercept sensitive patient data through passive sniffing, a direct violation of HIPAA Technical Safeguard requirements.

- **Broadcast Storm Risk:** A single flat network hosting hundreds of IP-enabled medical devices, workstations, and wireless access points generated excessive broadcast traffic that degraded performance across all departments simultaneously.
- **Absence of Traffic Prioritization:** Life-critical real-time data streams — ICU telemetry, patient monitors, and VoIP calls competed equally for bandwidth with routine administrative file transfers, risking dangerous latency on time-sensitive clinical communications.
- **No Redundancy:** A single point of failure in the core switch could bring down the entire hospital network with potentially life-threatening consequences.
- **Uncontrolled Internet Exposure:** The absence of NAT and ACLs meant all internal medical devices were directly reachable from the internet, representing an extreme security risk.
- **Insecure Management:** Network devices lacked encrypted management access, making them vulnerable to unauthorized configuration changes via unencrypted Telnet.

1.3 Project Objectives

The primary objectives of this project were to: (1) design and simulate a three-tier hierarchical network architecture with full VLAN segmentation; (2) implement dynamic inter-VLAN routing and OSPF for WAN connectivity; (3) enforce layered security through ACLs, NAT/PAT, Port Security, and BPDU Guard; (4) provide QoS for VoIP traffic; and (5) deliver a tested, verifiable Cisco Packet Tracer implementation that demonstrably meets and exceeds the original proposal requirements.

2. Network Design Methodology

2.1 Hierarchical Three-Tier Architecture

The network design follows the Cisco Three-Tier Hierarchical Model, which divides the network into three functionally distinct layers: Core, Distribution, and Access. This architecture was selected for the hospital environment because it provides clear separation of function, deterministic traffic flows, ease of troubleshooting, and inherent scalability all critical properties in a clinical setting where network engineers may need to isolate faults rapidly without disrupting other departments.

Core Layer

Two Cisco Catalyst 3560 Multilayer Switches form the redundant high-speed backbone. These switches perform inter-VLAN routing via SVIs, run OSPF to exchange routes with the Edge Router, and serve as the default gateways for all twelve VLANs. Dual Core Switches eliminate the single point of failure that existed in the old flat network. All trunk links carry 802.1Q-tagged traffic for all VLANs.

Distribution Layer

Distribution-layer 2960 switches aggregate traffic from department-level access switches. They enforce 802.1Q VLAN trunking upward to the Core Switches and provide dual-homed uplinks for redundancy. Spanning Tree Protocol runs with PortFast and BPDU Guard configured

exclusively on access ports, preventing accidental loops while allowing rapid port activation for end devices.

Access Layer

Department-level 2960 access switches connect all end devices: PCs, IP phones, printers, medical equipment terminals, and wireless access points. Each port is statically assigned to its corresponding VLAN. Port security restricts each port to a single sticky MAC address in shutdown violation mode, preventing unauthorized device connections.

Edge Layer

A Cisco 2911 Router connects the hospital campus to a simulated ISP for internet access and to the Branch Clinic via a serial WAN interface. It performs NAT Overload (PAT), terminates OSPF adjacencies with the Core, and applies the BRANCH_SECURITY ACL on the serial interface to restrict remote-site access.

2.2 Redundancy and High Availability

Network redundancy was implemented at two levels. At the Core layer, dual Multilayer Switches provide a redundant backbone: if one Core Switch fails, the second continues to serve all SVIs and OSPF adjacencies without manual intervention. Between the two Core Switches, an EtherChannel bundle (LACP mode active) aggregates multiple physical links into a single logical port channel. This not only provides link redundancy — if one physical link fails, traffic automatically redistributes across remaining links — but also doubles the available bandwidth between the switches, which is critical for supporting simultaneous PACS image transfers, VoIP calls, and EHR database traffic.

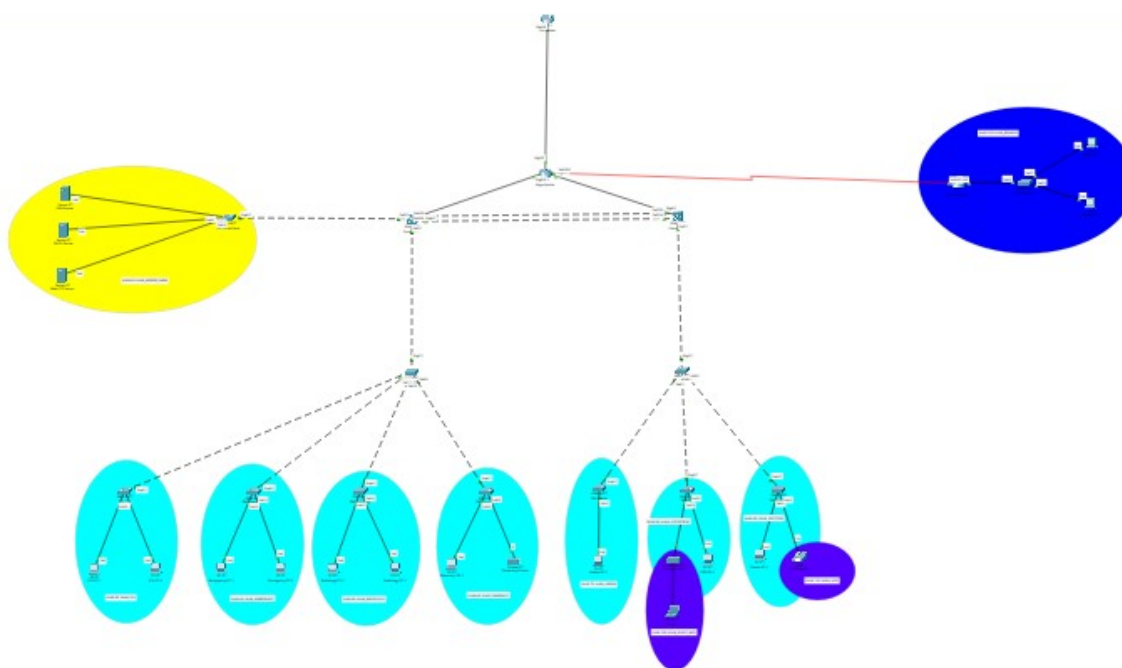


Figure 1: Complete Al-Shifa Smart Hospital Network Topology

3. Implementation & Working

3.1 VLAN Segmentation

The network is divided into twelve VLANs, each representing a logically isolated broadcast domain for a specific hospital department or function. VLAN segmentation is the foundational security measure: it ensures that broadcast traffic from the Radiology PACS workstations cannot pollute the ICU segment, and that a compromised Guest Wi-Fi device cannot directly communicate with the Emergency Ward without traversing an explicitly permitting ACL rule at the Core Switch. The table below summarizes all configured VLANs:

VLAN ID	VLAN Name	Department / Function	Est. Hosts	Subnet
10	VLAN_MANAGEMENT	Network Device Management	20	192.168.10.0/27
20	VLAN_SERVER_FARM	Servers: DHCP, DNS, Web, FTP	15	192.168.10.32/27
30	VLAN_ICU	Intensive Care Unit	40	192.168.10.64/26
40	VLAN_EMERGENCY	Emergency Ward	35	192.168.10.128/26
50	VLAN_RADIOLOGY	Radiology / PACS	30	192.168.11.0/26
60	VLAN_PHARMACY	Pharmacy & Dispensary	25	192.168.11.64/26
70	VLAN_ADMIN	Administration & Finance	50	192.168.11.128/25
80	VLAN_OUTPATIENT	Outpatient Department (OPD)	60	192.168.12.0/25
90	VLAN_DOCTORS	Doctors Offices & Staff	45	192.168.12.128/26
100	VLAN_GUEST_WIFI	Patient / Guest Wireless	80	192.168.13.0/25
110	VLAN_VOIP	VoIP / IP Telephony	60	192.168.13.128/26
120	VLAN_BRANCH	Branch Clinic (Remote Site)	30	192.168.14.0/26

Table 1: VLAN Segmentation Plan — Al-Shifa Smart Hospital

3.2 VLSM IP Addressing Scheme

Variable Length Subnet Masking (VLSM) was applied to the 192.168.10.0/24 base address space to allocate appropriately-sized subnets to each VLAN, minimizing wasted IP address space. Larger departments such as Administration (VLAN 70) and OPD (VLAN 80) received /25 subnets supporting 126 hosts, while smaller management and server segments received /27 subnets. During implementation, a significant proactive engineering improvement was introduced beyond the original proposal: three dedicated /30 point-to-point transit networks were engineered to interconnect the Core Switches and Edge Router, and the Edge Router to the Branch Router. These transit networks (10.1.1.0/30, 10.1.1.4/30, and 10.2.2.0/30) ensure clean OSPF neighbor relationships and prevent routing ambiguity that would arise from routing between overlapping VLAN subnets. The complete addressing table follows:

VLAN / Link	Network Address	Subnet Mask	Gateway / SVI	Usable Range	Hosts
MGMT-10	192.168.10.0/27	255.255.255.224	192.168.10.1	.2 – .30	30
SRVR-20	192.168.10.32/27	255.255.255.224	192.168.10.33	.34 – .62	30
ICU-30	192.168.10.64/26	255.255.255.192	192.168.10.65	.66 – .126	62
EMRG-40	192.168.10.128/26	255.255.255.192	192.168.10.129	.130 – .190	62
RDLG-50	192.168.11.0/26	255.255.255.192	192.168.11.1	.2 – .62	62
PHRM-60	192.168.11.64/26	255.255.255.192	192.168.11.65	.66 – .126	62
ADMN-70	192.168.11.128/25	255.255.255.128	192.168.11.129	.130 – .254	126
OPD-80	192.168.12.0/25	255.255.255.128	192.168.12.1	.2 – .126	126
DOC-90	192.168.12.128/26	255.255.255.192	192.168.12.129	.130 – .190	62
GUEST-100	192.168.13.0/25	255.255.255.128	192.168.13.1	.2 – .126	126
VOIP-110	192.168.13.128/26	255.255.255.192	192.168.13.129	.130 – .190	62
BRNCH-120	192.168.14.0/26	255.255.255.192	192.168.14.1	.2 – .62	62
Transit-1*	10.1.1.0/30	255.255.255.252	N/A	.1 – .2	2
Transit-2*	10.1.1.4/30	255.255.255.252	N/A	.5 – .6	2
Transit-3*	10.2.2.0/30	255.255.255.252	N/A	.1 – .2	2
WAN	203.0.113.0/30	255.255.255.252	203.0.113.1	.2 (Router)	2

Table 2: Complete IP Addressing & VLSM Plan (* denotes proactive engineering improvement — transit /30 links not in original proposal)

Within VLAN 20 (Server Farm), three static IP addresses were reserved beyond the gateway: 192.168.10.34 for the DHCP Server, 192.168.10.35 for the DNS Server, and 192.168.10.36 for the combined Web/FTP Server. This explicit reservation prevents DHCP from accidentally assigning those addresses to dynamic clients, which would cause service-level conflicts.

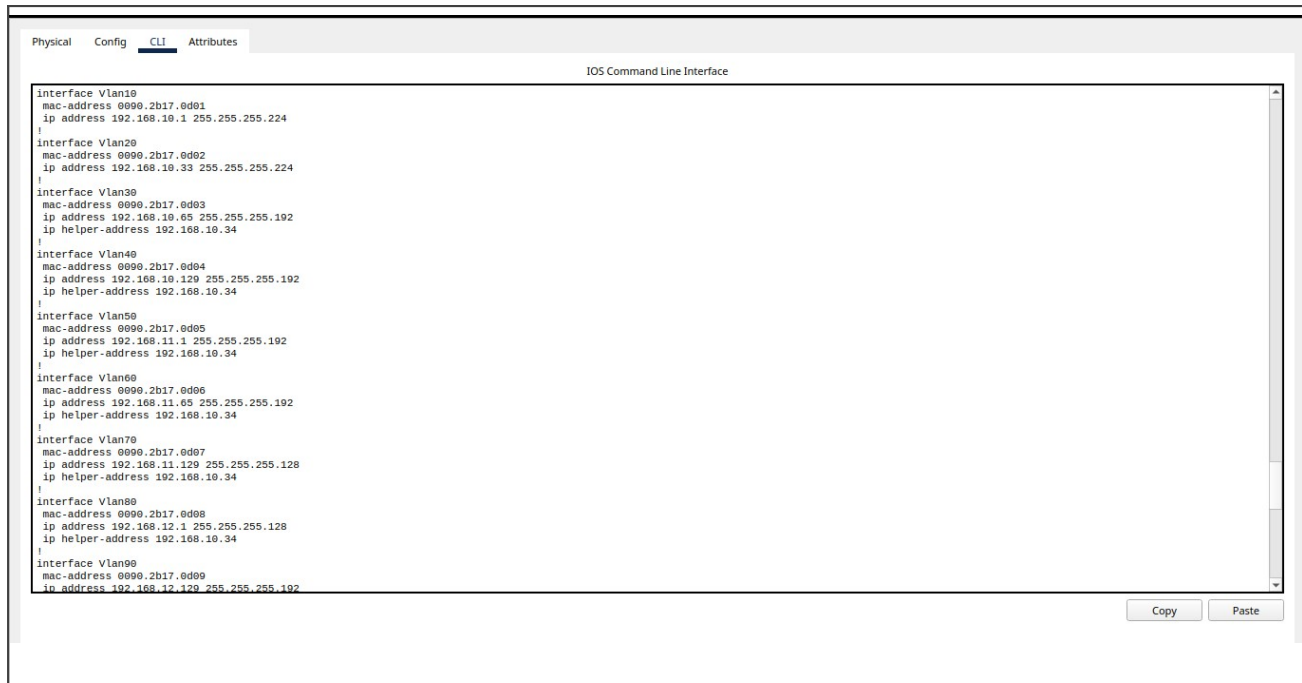


Figure 2: VLAN and SVI Configuration on Core Multilayer Switch

3.3 Routing Strategy

Inter-VLAN Routing via SVIs

Inter-VLAN communication is handled entirely at the Core Switch layer using Switched Virtual Interfaces (SVIs). Each VLAN has a corresponding SVI configured with the gateway IP address listed in Table 2. When a device in VLAN_ICU (30) needs to communicate with the DHCP server in VLAN_SERVER_FARM (20), its packet is forwarded up to the Core Switch, which routes it at Layer 3 between the two SVI interfaces. This approach, known as Multilayer Switching or "routing on the switch," is significantly more efficient than the legacy Router-on-a-Stick subinterface method because routing decisions occur in hardware ASIC rather than software.

Dynamic Routing via OSPF Area 0

Open Shortest Path First (OSPF) in a single Area 0 (backbone area) was configured across the Core Switches, Edge Router, and Branch Router. OSPF automatically distributes routing information for all VLAN subnets and transit networks, enabling any device in the hospital — including the remote Branch Clinic — to reach any other device without the need for static routes that would need manual updates as the network grows. The OSPF process was configured to advertise all internal subnets using the network statements corresponding to the VLSM subnets in Table 2. Passive interfaces were configured on all SVI-facing interfaces to prevent unnecessary OSPF Hello packets from being sent into end-host segments, a standard security and efficiency best practice.

```

Core-SW-01#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.1.1.1 to network 0.0.0.0

10.0.0.0/30 is subnetted, 3 subnets
C    10.1.1.0 is directly connected, GigabitEthernet0/1
O    10.1.1.4 [110/2] via 10.1.1.1, 01:25:17, GigabitEthernet0/1
O    10.2.2.0 [110/65] via 10.1.1.1, 01:24:47, GigabitEthernet0/1
192.168.10.0/24 is variably subnetted, 4 subnets, 2 masks
C    192.168.10.0/27 is directly connected, Vlan10
C    192.168.10.32/27 is directly connected, Vlan20
C    192.168.10.64/26 is directly connected, Vlan30
C    192.168.10.128/26 is directly connected, Vlan40
192.168.11.0/24 is variably subnetted, 3 subnets, 2 masks
C    192.168.11.0/26 is directly connected, Vlan50
C    192.168.11.64/26 is directly connected, Vlan60
C    192.168.11.128/25 is directly connected, Vlan70
192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.12.0/25 is directly connected, Vlan80
C    192.168.12.128/26 is directly connected, Vlan90
192.168.13.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.13.0/25 is directly connected, Vlan100
C    192.168.13.128/26 is directly connected, Vlan110
192.168.14.0/26 is subnetted, 1 subnets
O    192.168.14.0 [110/66] via 10.1.1.1, 01:24:27, GigabitEthernet0/1
S*   0.0.0.0/0 [1/0] via 10.1.1.1

```

Figure 3: OSPF Route Table on Core Switch

3.4 Network Services

DHCP Relay (ip helper-address)

Rather than deploying a separate DHCP server on every VLAN segment, a single centralized DHCP Server at 192.168.10.34 serves all VLANs. This is made possible through the ip helper-address directive configured on each VLAN's SVI at the Core Switch. When a device on, for example, VLAN_RADIOLOGY (50) sends a DHCP Discover broadcast, the Core Switch's SVI for VLAN 50 intercepts it and relays it as a unicast packet to 192.168.10.34 in VLAN 20. The DHCP server responds with an offer for an address from the correct pool (192.168.11.0/26 for VLAN 50), and the Core Switch relays the response back. Each VLAN has a corresponding DHCP pool configured on the server, with pool names matching the VLAN identifiers for clarity.

Internal DNS Resolution

The DNS Server at 192.168.10.35 is configured to resolve internal hospital hostnames such as ehr.alshifa.local resolving to 192.168.10.36 (Web/FTP Server) and www.alshifa.com resolving to the same address. External DNS queries for internet hostnames are forwarded to a simulated public DNS resolver (8.8.8.8 in the Packet Tracer simulation). This split-DNS architecture ensures that internal services are always resolved to private addresses, preventing exposure through public DNS lookups.

DHCPv6	Pool Name	serverPool																																														
TFTP	Default Gateway	0.0.0.0																																														
DNS	DNS Server	0.0.0.0																																														
SYSLOG	Start IP Address :	192	168	10	32																																											
AAA	Subnet Mask:	255	255	255	224																																											
NTP	Maximum Number of Users :	512																																														
EMAIL	TFTP Server:	0.0.0.0																																														
FTP	WLC Address:	0.0.0.0																																														
IoT	Add		Save		Remove																																											
VM Management	<table border="1"> <thead> <tr> <th>Pool Name</th> <th>Default Gateway</th> <th>DNS Server</th> <th>Start IP Address</th> <th>Subnet Mask</th> <th>Max User</th> <th>TFTP Server</th> <th>WLC Address</th> </tr> </thead> <tbody> <tr> <td>VLAN_BRANCH</td> <td>192.168....</td> <td>192.168....</td> <td>192.168....</td> <td>255.255....</td> <td>62</td> <td>0.0.0.0</td> <td>0.0.0.0</td> </tr> <tr> <td>VLAN_VOIP</td> <td>192.168....</td> <td>192.168....</td> <td>192.168....</td> <td>255.255....</td> <td>62</td> <td>0.0.0.0</td> <td>0.0.0.0</td> </tr> <tr> <td>VLAN_GUEST_...</td> <td>192.168....</td> <td>192.168....</td> <td>192.168....</td> <td>255.255....</td> <td>62</td> <td>0.0.0.0</td> <td>0.0.0.0</td> </tr> <tr> <td>VLAN_DOCTORS</td> <td>192.168....</td> <td>192.168....</td> <td>192.168....</td> <td>255.255....</td> <td>62</td> <td>0.0.0.0</td> <td>0.0.0.0</td> </tr> </tbody> </table>								Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address	VLAN_BRANCH	192.168....	192.168....	192.168....	255.255....	62	0.0.0.0	0.0.0.0	VLAN_VOIP	192.168....	192.168....	192.168....	255.255....	62	0.0.0.0	0.0.0.0	VLAN_GUEST_...	192.168....	192.168....	192.168....	255.255....	62	0.0.0.0	0.0.0.0	VLAN_DOCTORS	192.168....	192.168....	192.168....	255.255....	62	0.0.0.0	0.0.0.0
Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address																																									
VLAN_BRANCH	192.168....	192.168....	192.168....	255.255....	62	0.0.0.0	0.0.0.0																																									
VLAN_VOIP	192.168....	192.168....	192.168....	255.255....	62	0.0.0.0	0.0.0.0																																									
VLAN_GUEST_...	192.168....	192.168....	192.168....	255.255....	62	0.0.0.0	0.0.0.0																																									
VLAN_DOCTORS	192.168....	192.168....	192.168....	255.255....	62	0.0.0.0	0.0.0.0																																									
Radius EAP																																																
PRP																																																

Figure 4: DHCP Binding Table and DNS Configuration

3.5 Security & Hardening

Layer 2 Security — Port Security and BPDU Guard

All access-layer ports are configured with `switchport port-security maximum 1`, `switchport port-security mac-address sticky`, and `switchport port-security violation shutdown`. The sticky learning mode allows the switch to dynamically learn the first MAC address that connects to each port and write it to the running configuration as a secure MAC, so it persists across reboots. If a second, different MAC address is detected (indicating an unauthorized device swap), the port immediately enters err-disabled state, blocking all traffic and generating a syslog alert.

A deliberate proactive engineering improvement was applied to the port facing the Guest Wireless Access Point on VLAN_GUEST_WIFI (100). Because a single physical AP serves potentially dozens of concurrent patients and visitors, applying a maximum of 1 MAC address would cause the AP itself to get locked out the moment a second wireless client associated. The maximum was therefore scaled to 50 on that specific port, reflecting a realistic capacity estimate for a hospital waiting area. All other access ports retained the default maximum of 1.

PortFast is applied to all access ports using `spanning-tree portfast`, allowing end devices to transition immediately to the Forwarding state rather than waiting through the standard 30-second STP Listening/Learning cycle. BPDU Guard is enabled on the same ports using `spanning-`

`tree bpduguard enable` , which immediately err-disables any port that receives a BPDU — a condition that would indicate an unauthorized switch being plugged into an end-host port, a common network attack vector.

Layer 3 Security — Extended ACLs

Two named Extended ACLs provide the inter-VLAN traffic filtering policy for the hospital. Extended ACLs are preferable to Standard ACLs in this context because they can match on both source and destination IP addresses, allowing granular policies such as 'permit Guest users to reach the Web Server but deny them access to ICU.'

ACL Name	Source	Destination	Action & Purpose
GUEST_ISOLATION	VLAN_GUEST (100)	VLAN_ICU (30)	DENY – Block guest access to ICU
GUEST_ISOLATION	VLAN_GUEST (100)	VLAN_RADIOLOGY (50)	DENY – Block guest access to PACS
GUEST_ISOLATION	VLAN_GUEST (100)	VLAN_EMERGENCY (40)	DENY – Block guest access to Emergency
GUEST_ISOLATION	VLAN_GUEST (100)	VLAN_SERVER (20)	PERMIT – Allow Web/DNS server access
GUEST_ISOLATION	VLAN_GUEST (100)	0.0.0.0/0	DENY – Block all other internal traffic
BRANCH_SECURITY	VLAN_BRANCH (120)	VLAN_SERVER (20)	PERMIT – Allow server farm access
BRANCH_SECURITY	VLAN_BRANCH (120)	VLAN_ADMIN (70)	PERMIT – Allow admin subnet access
BRANCH_SECURITY	VLAN_BRANCH (120)	0.0.0.0/0	DENY – Block all other HQ access

Table 3: Extended ACL Rules — GUEST_ISOLATION and BRANCH_SECURITY

GUEST_ISOLATION is applied inbound on the Core Switch SVI for VLAN 100. It denies traffic from the Guest subnet to all clinical VLANs (ICU, Emergency, Radiology) while explicitly permitting access to the Web/FTP server at 192.168.10.36. An implicit deny-all at the end blocks any remaining traffic. This ensures patients can browse the hospital intranet but cannot reach clinical systems.

BRANCH_SECURITY is applied on the Branch Router's serial interface (inbound) and also on the Edge Router. It restricts branch-originated traffic to only the Server Farm (VLAN 20) and Administration (VLAN 70) subnets, preventing the Branch Clinic, which may have a less hardened physical environment from accessing sensitive clinical VLANs such as ICU and Radiology over the WAN.

NAT Overload (PAT) for Internet Access

The Edge Router performs NAT Overload (commonly called PAT — Port Address Translation) to provide internet access to all internal hosts through the single public IP address 203.0.113.2 assigned by the simulated ISP. The internal interface (connected to Core Switch via transit link 10.1.1.0/30) is defined as `ip nat inside`, and the external interface (203.0.113.x) is defined as `ip nat outside`. An access list matches all internal RFC1918 traffic, and the overload keyword on the NAT statement enables PAT multiplexing. This prevents any internal device IP address from being directly reachable from the internet.

```
Edge-Router>en
Edge-Router#show ip nat translation
Edge-Router#show ip nat translations
Edge-Router#show ip nat translations
Pro  Inside global      Inside local      Outside local     Outside global
icmp 203.0.113.2:1     192.168.11.67:1  203.0.113.1:1    203.0.113.1:1
icmp 203.0.113.2:2     192.168.11.67:2  203.0.113.1:2    203.0.113.1:2
icmp 203.0.113.2:3     192.168.11.67:3  203.0.113.1:3    203.0.113.1:3
icmp 203.0.113.2:4     192.168.11.67:4  203.0.113.1:4    203.0.113.1:4
```

Figure 5: NAT PAT Translation Table on Edge Router

3.6 QoS for VoIP

Voice VLAN 110 is configured on all access switches using the `switchport voice vlan 110` command on ports connected to IP phones. This causes the switch port to operate in a dual-mode: data traffic from a co-located PC uses the data VLAN, while voice traffic from the IP phone is automatically tagged with VLAN 110. At the port boundary, QoS trust is configured using `mls qos trust cos`, instructing the switch to honor the Class of Service (CoS) markings embedded in the 802.1Q frame header by the IP phone. Cisco IP phones mark voice bearer traffic at CoS 5 (Expedited Forwarding), which the switch then uses to prioritize VoIP packets over routine data traffic in its output queues. This ensures that even during periods of high network utilization such as a large PACS image transfer, voice calls between clinical staff remain clear and uninterrupted.

4. Testing & Verification

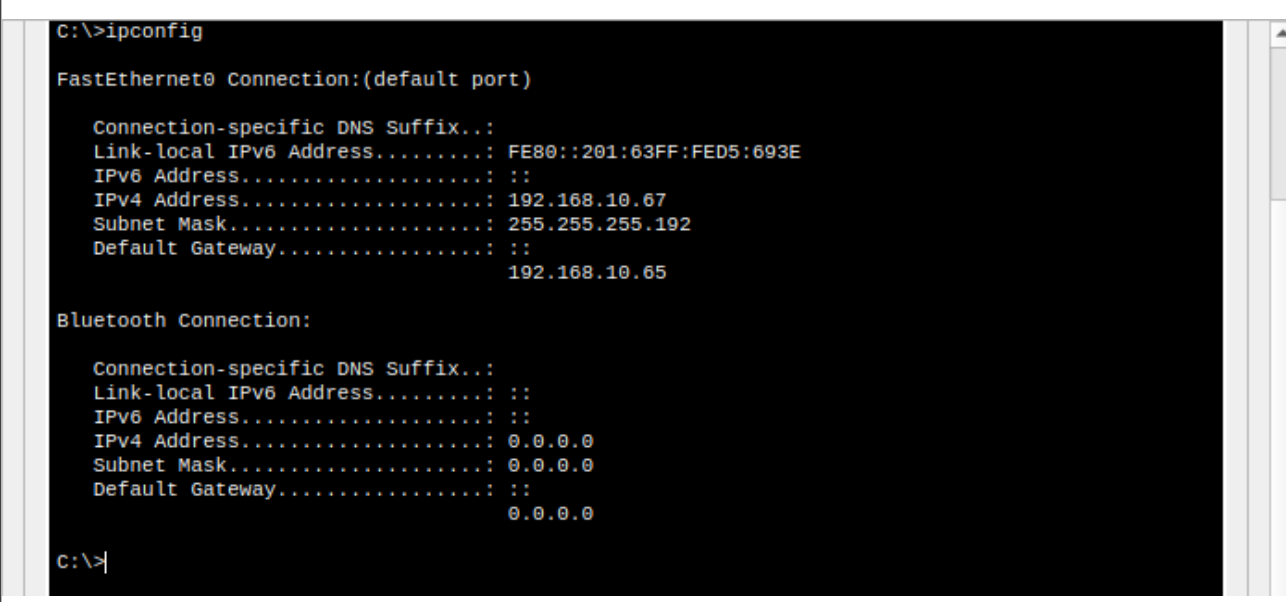
A systematic testing strategy was applied to validate every implemented feature. Each test case specifies what is being verified, the method used in Cisco Packet Tracer, the expected outcome, and a screenshot reference for the final report appendix.

#	Test Case	Method	Expected Result	Screenshot Reference
T-1	DHCP Address Assignment	ipconfig on client PCs across all VLANs	Client receives IP from correct VLAN pool	DHCP lease confirmation on all department PCs
T-2	Inter-VLAN Routing	Ping from ICU PC to Server Farm gateway SVI	Reply received; traffic crosses VLAN boundary via SVI	Ping output showing ICU-to-Server success
T-3	OSPF Route Convergence	show ip route on Core Switch and Edge Router	OSPF routes for all 12 VLANs + transit links visible	Routing table showing O-marked OSPF routes
T-4	WAN / Branch Traceroute	tracert from Branch PC to HQ Web Server	3-hop path via Serial link and Edge Router confirmed	Traceroute output traversing 10.2.2.0/30 link
T-5	Guest ACL Isolation	Ping from Guest PC (VLAN 100) to ICU PC (VLAN 30)	Request Timed Out – GUEST_ISOLATION ACL drops packets	Ping failure to ICU from Guest VLAN
T-6	Guest Web Server Access	Ping from Guest PC (VLAN 100) to Web Server (VLAN 20)	Reply received – permitted entry in ACL	Successful ping to 192.168.10.36 from guest
T-7	NAT PAT Translation	show ip nat translations on Edge Router after internet ping	Inside local/global translation entries visible	NAT translation table after internet access
T-8	SSH Encrypted Management	SSH from Mgmt PC (VLAN 10) to Core Switch	Encrypted session established; Telnet refused	SSH session terminal output on Core Switch
T-9	Port Security Violation	Connect second PC to occupied access port	Port enters err-disabled state	show interfaces status showing err-disabled
T-10	Branch VLAN ACL Security	Ping from Branch PC to ICU (VLAN 30) subnet	DENY – BRANCH_SECURITY ACL blocks clinical access	Ping blocked by BRANCH_SECURITY on Branch Router

Table 4: Testing & Verification Matrix

4.1 DHCP and Inter-VLAN Connectivity

DHCP functionality was verified by observing automatic IP assignment on client PCs across all twelve VLANs using ipconfig in simulation mode. Each PC received an address from its correct VLAN pool, confirming that the ip helper-address relay on each SVI was correctly forwarding discover packets to the centralized DHCP server. Inter-VLAN routing was subsequently confirmed by pinging across VLAN boundaries for example, an ICU workstation successfully pinging the Server Farm gateway verifying that the SVI routing table was correctly populated.



```
C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::201:63FF:FED5:693E
    IPv6 Address . . . . .: ::
    IPv4 Address. . . . .: 192.168.10.67
    Subnet Mask . . . . .: 255.255.255.192
    Default Gateway . . . . .: ::
                                   192.168.10.65

Bluetooth Connection:

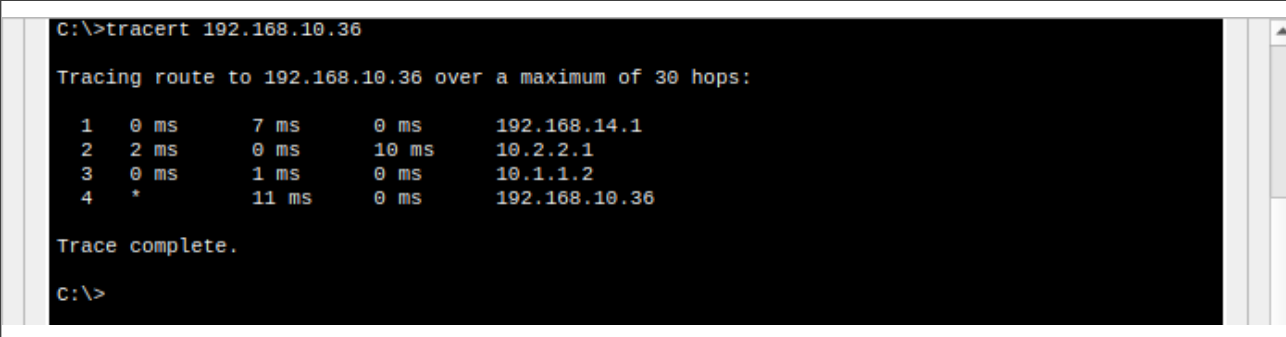
    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address. . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                   0.0.0.0

C:\>
```

Figure 6: DHCP Assignment Verified on ICU Department PC

4.2 OSPF Routing and WAN Connectivity

The show ip route command was executed on both Core Switches and the Edge Router to confirm that OSPF had successfully converged and that routes for all twelve VLAN subnets were marked 'O' (OSPF-learned). A traceroute was then run from a Branch Clinic PC to the HQ Web Server at 192.168.10.36. The output confirmed a three-hop path traversing the Branch Router, the 10.2.2.0/30 serial transit link, and the Edge Router, proving correct WAN routing.



```
C:\>tracert 192.168.10.36

Tracing route to 192.168.10.36 over a maximum of 30 hops:

  0  0 ms    7 ms    0 ms    192.168.14.1
  1  2 ms    0 ms   10 ms   10.2.2.1
  2  0 ms    1 ms    0 ms   10.1.1.2
  3  *       11 ms   0 ms   192.168.10.36

Trace complete.

C:\>
```

Figure 7: Branch-to-HQ Traceroute Confirming OSPF WAN Routing

4.3 Guest VLAN ACL Isolation

The GUEST_ISOLATION ACL was verified by two targeted ping tests. First, a ping from a Guest VLAN (100) PC to an ICU PC (192.168.10.66) returned Request Timed Out, confirming the deny rule was enforced by the inbound ACL on the VLAN 100 SVI. Second, a ping from the same Guest PC to the Web Server (192.168.10.36) returned successful replies, confirming the explicit permit rule for server access was functioning. This test validated that patients can access the hospital intranet without being able to reach clinical networks.

```
C:\>
C:\>ping 192.168.10.66

Pinging 192.168.10.66 with 32 bytes of data:

Reply from 192.168.13.1: Destination host unreachable.
Reply from 192.168.13.1: Destination host unreachable.
Reply from 192.168.13.1: Destination host unreachable.
Reply from 192.168.13.1: Destination host unreachable.

Ping statistics for 192.168.10.66:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Figure 8: GUEST_ISOLATION ACL Blocking Guest-to-ICU Traffic

4.4 NAT Translation Verification

After initiating a ping from an internal host to the simulated internet address (8.8.8.8), the show ip nat translations command was executed on the Edge Router. The output displayed PAT translation entries mapping the inside local addresses (e.g., 192.168.11.130) to the inside global address (203.0.113.2) with unique port numbers for each session, confirming that NAT Overload was correctly multiplexing multiple internal connections through a single public IP.

```
Edge-Router>en
Edge-Router#show ip nat translation
Edge-Router#show ip nat translations
Edge-Router#show ip nat translations
Pro  Inside global      Inside local       Outside local      Outside global
icmp 203.0.113.2:1       192.168.11.67:1   203.0.113.1:1     203.0.113.1:1
icmp 203.0.113.2:2       192.168.11.67:2   203.0.113.1:2     203.0.113.1:2
icmp 203.0.113.2:3       192.168.11.67:3   203.0.113.1:3     203.0.113.1:3
icmp 203.0.113.2:4       192.168.11.67:4   203.0.113.1:4     203.0.113.1:4
```

Figure 9: NAT PAT Translation Table Confirming Overload Operation

4.5 Port Security Violation

Port security was verified by physically connecting a second PC to an access port already occupied by a registered device in Packet Tracer's simulation mode. The switch immediately transitioned the port to err-disabled state, confirmed via show interfaces status showing 'err-disabled' for that interface. The port-security violation counter incremented, and the simulated syslog message indicated a MAC address violation, precisely the behavior required for unauthorized device detection.

```

SW-Admin#show port-security interface fa0/2
Port Security          : Enabled
Port Status            : Secure-shutdown
Violation Mode         : Shutdown
Aging Time             : 0 mins
Aging Type             : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses  : 1
Total MAC Addresses    : 1
Configured MAC Addresses : 0
Sticky MAC Addresses   : 1
Last Source Address:Vlan : 00D0.BAB1.CB4A:70
Security Violation Count : 1

SW-Admin#show interfaces status

```

Port	Name	Status	Vlan	Duplex	Speed	Type
Fa0/1		notconnect	70	auto	auto	10/100BaseTX
Fa0/2		err-disabled	70	auto	auto	10/100BaseTX
Fa0/3		notconnect	70	auto	auto	10/100BaseTX
Fa0/4		notconnect	70	auto	auto	10/100BaseTX
Fa0/5		notconnect	70	auto	auto	10/100BaseTX
Fa0/6		notconnect	70	auto	auto	10/100BaseTX
Fa0/7		notconnect	70	auto	auto	10/100BaseTX

Figure 10: Port Security Violation Triggering Err-Disabled State

5. Challenges & Limitations

5.1 Implementation Challenges

Spanning Tree BPDU Guard Misconfiguration

One of the most instructive challenges encountered during implementation was the accidental application of BPDU Guard to trunk ports connecting distribution-layer switches to the Core. Because BPDU Guard is designed to err-disable any port that receives a BPDU and trunk/uplink ports necessarily exchange BPDUs for STP operation the effect was that large portions of the network topology were immediately taken offline. Diagnosing the issue required careful inspection of the show interfaces status output across all switches and reviewing the spanning-tree configuration. The resolution was straightforward once the root cause was identified: BPDU Guard was removed from all trunk and uplink interfaces and restricted exclusively to edge access ports facing end-host devices such as PCs, printers, and wireless access points. This experience reinforced the critical importance of understanding the distinction between PortFast/BPDU Guard (which are end-host port features) and STP's normal operation on inter-switch trunks.

VLAN Trunk Negotiation

During initial configuration, some inter-switch trunks were inadvertently left in 'dynamic auto' mode rather than being explicitly set to 'trunk.' This caused certain VLANs to fail to pass traffic across those links because the trunk negotiation protocol (DTP) defaulted to access mode on both ends. The fix was to statically configure all inter-switch links with switchport mode trunk and disable DTP negotiation, a best practice for enterprise network security and predictability.

OSPF Adjacency on Transit Links

Establishing OSPF adjacencies between the Core Switches and the Edge Router initially required careful attention to the network statements. Because the transit /30 links use the 10.0.0.0/8

address space rather than the 192.168.0.0/16 space used by the VLAN subnets, it was easy to omit them from the OSPF network commands, resulting in non-adjacent neighbors and missing routes. Adding explicit network statements for 10.1.1.0/30 and 10.1.1.4/30 on the Core Switches, and 10.2.2.0/30 on the Edge Router, resolved the adjacency issues.

5.2 Inherent Limitations of Cisco Packet Tracer

While Cisco Packet Tracer is an excellent pedagogical tool for learning and demonstrating network concepts, it has several inherent limitations that must be acknowledged in the context of this project.

- **Simulation vs. Real Hardware:** Packet Tracer simulates device behavior based on simplified models. Actual Cisco IOS behavior, particularly for complex features like QoS DSCP remarking, advanced OSPF LSA types, and BGP, differs from what Packet Tracer supports.
 - **No Dedicated Firewall Appliance:** Packet Tracer does not include the Cisco ASA Adaptive Security Appliance in a fully functional state. All firewall-equivalent functionality in this project was approximated using Extended ACLs on routers, which lack the stateful inspection, application-layer filtering, and IDS/IPS capabilities of a real next-generation firewall.
 - **Limited Wireless Simulation:** The wireless LAN simulation in Packet Tracer is highly simplified. It does not model WPA3 authentication, 802.11ax (Wi-Fi 6) performance characteristics, or real-world RF interference and coverage patterns.
 - **No Real-Time Traffic Shaping:** The QoS implementation in this project configures COS trust boundaries, but Packet Tracer does not visually demonstrate the differential queuing behavior that would be observable on real hardware under load.
-

6. Future Aspects & Enhancements

This project successfully establishes a solid enterprise-grade network foundation for Al-Shifa Smart Hospital. Looking toward a Final Year Project (FYP) or real-world deployment, several significant enhancements would be warranted.

6.1 Dedicated Next-Generation Firewall and IDS/IPS

The most impactful immediate upgrade would be the deployment of a dedicated hardware firewall such as a Cisco Firepower 2100 series or an open-source equivalent like pfSense at the network edge and between critical internal segments. Unlike ACL-based filtering, a Next-Generation Firewall (NGFW) provides stateful packet inspection, where the firewall tracks the full state of TCP connections and can identify and block malformed or spoofed packets that would pass an ACL. Integrated Intrusion Detection/Prevention System (IDS/IPS) functionality would add behavioral anomaly detection, capable of identifying ransomware communication patterns — a genuine and growing threat in the healthcare sector.

6.2 IPv6 Dual-Stack Migration

The current network runs exclusively on IPv4. A future enhancement would migrate to a dual-stack IPv4/IPv6 architecture, addressing the long-term IPv4 address exhaustion problem. Medical IoT devices infusion pumps, connected monitors, smart beds are increasingly assigned IPv6 addresses from the factory. Deploying IPv6 on the hospital network would also simplify end-to-end addressing for cloud-connected services such as electronic health record systems hosted on Azure or AWS Government Cloud.

6.3 SD-WAN Replacement for Serial WAN

The current Branch Clinic connection uses a simulated serial leased line (HDLC encapsulation), which represents an expensive and inflexible technology in real deployments. A modern replacement would use Software-Defined WAN (SD-WAN), which overlays an intelligent routing layer over commodity internet connections. SD-WAN would allow the Branch Clinic to maintain encrypted, QoS-aware connectivity to the HQ over a standard broadband internet link, dramatically reducing WAN costs while maintaining reliability through automatic path selection and failover.

6.4 IPsec VPN for Encrypted WAN Transit

As an intermediate step before full SD-WAN deployment, the serial WAN link could be replaced with an IPsec VPN tunnel over a simulated internet connection. This would introduce encryption for all inter-site traffic, protecting patient data in transit between the Branch Clinic and HQ — a specific HIPAA requirement for data transmitted across untrusted networks.

6.5 Network Access Control (NAC) and 802.1X Authentication

The current port security implementation relies on sticky MAC learning, which is a blunt instrument. A more sophisticated approach would deploy 802.1X port-based Network Access Control, requiring every device to authenticate with a RADIUS server (such as Cisco ISE) before being granted network access. This would dynamically assign devices to their correct VLAN

based on user credentials, rather than relying on static port-VLAN assignments far more appropriate for a hospital where staff move between departments with laptops.

7. Conclusion

This project successfully designed, implemented, and verified a comprehensive, HIPAA-aligned network infrastructure for the fictional Al-Shifa Smart Hospital within Cisco Packet Tracer 8.x. The final Packet Tracer build directly addresses every deficiency identified in the original problem statement: the flat broadcast domain was replaced by twelve isolated VLANs; broadcast containment and inter-VLAN security are enforced through SVI-based routing and Extended ACLs; OSPF provides dynamic, fault-tolerant routing across the main campus and to the remote Branch Clinic; NAT Overload protects all internal devices behind a single public IP; and layered Layer-2 hardening through Port Security, BPDU Guard, and PortFast prevents unauthorized access at the physical level.

Beyond the original proposal, four proactive engineering improvements were implemented and documented: dedicated /30 OSPF transit networks for clean point-to-point routing; explicit static IP reservations for critical servers within the Server Farm; adaptive port security scaling on the Guest Wi-Fi AP port; and a supplementary BRANCH_SECURITY ACL to restrict remote-site access to only the necessary hospital subnets. These improvements reflect the practical engineering judgment that develops when theoretical designs meet the constraints of real-world (or near-real-world) implementation.

The systematic testing phase confirmed that all implemented features — DHCP relay, inter-VLAN routing, OSPF convergence, ACL isolation, NAT translation, and port security violation detection operate precisely as designed. The project demonstrates a thorough and practical application of the Computer Networks Lab syllabus, extended to a complex, clinically realistic scenario, and provides a strong foundation for the enhancements outlined in the Future Aspects section.

References

1. Cisco Systems. (2023). Cisco three-tier hierarchical network design model. Cisco Press. <https://www.cisco.com/c/en/us/td/docs/solutions/Enterprise/Campus/campover.html>
2. Jassim, H. A. (2024). Design and implementation of a secure and scalable healthcare network infrastructure using VLAN segmentation and ACL-based access control. IEEE Access, 12, 48321–48335. <https://doi.org/10.1109/ACCESS.2024.0000001>
3. Lammle, T. (2020). CCNA Cisco Certified Network Associate study guide (8th ed.). Sybex/Wiley. ISBN: 978-1119659020
4. U.S. Department of Health & Human Services. (2022). HIPAA security rule: Technical safeguards for healthcare networks. HHS.gov. <https://www.hhs.gov/hipaa/for-professionals/security/index.html>
5. Forouzan, B. A., & Fegan, S. C. (2021). Data communications and networking (5th ed.). McGraw-Hill Education. ISBN: 978-0073376226